Neural Computation -- 1993

74 article(s), all issues

1. Conversion of Temporal Correlations Between Stimuli to Spatial Correlations Between Attractors

Meir Griniasty, Misha Tsodyks, Daniel J. Amit

https://doi.org/10.1162/neco.1993.5.1.1 | pp. 1-17

It is shown that a simple modification of synaptic structures (of the Hopfield type) constructed to produce autoassociative attractors, produces neural networks whose attractors are correlated with several (learned) patterns used in the construction of the matrix. The modification stores in the matrix a fixed sequence of uncorrelated patterns. The network then has correlated attractors, provoked by the uncorrelated stimuli. Thus, the network converts the temporal order (or temporal correlation) expressed by the sequence of patterns, into spatial correlations expressed in the distributions of neural activities in attractors. The model captures phenomena observed in single electrode recordings in performing monkeys by Miyashita et al. The correspondence is as close as to reproduce the fact that given uncorrelated patterns as sequentially learned stimuli, the attractors produced are significantly correlated up to a separation of 5 (five) in the sequence. This number 5 is universal in a range of parameters, and requires essentially no tuning. We then discuss learning scenarios that could lead to this synaptic structure as well as experimental predictions following from it. Finally, we speculate on the cognitive utility of such an arrangement.

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2. On the Realization of a Kolmogorov Network

Ji-Nan Lin, R. Unbehauen

https://doi.org/10.1162/neco.1993.5.1.18 | pp. 18-20

January 01 1993 On the Realization of a Kolmogorov Network In Special Collection: CogNet Ji-Nan Lin, Ji-Nan Lin Lehrstuhl für Allgemeine und Theoretische Elektrotechnik, Universität Erlangen-Nürnberg, Cauerstrasse 7, D-8520 Erlangen, Germany Search for other works by this author on: This Site Google Scholar Rolf Unbehauen Rolf Unbehauen Lehrstuhl für Allgemeine und Theoretische Elektrotechnik, Universität Erlangen-Nürnberg, Cauerstrasse 7, D-8520 Erlangen, Germany Search for other works by this author on: This Site Google Scholar Author and Article Information Ji-Nan Lin Lehrstuhl für Allgemeine und Theoretische Elektrotechnik, Universität Erlangen-Nürnberg, Cauerstrasse 7, D-8520 Erlangen, Germany Rolf Unbehauen Lehrstuhl für Allgemeine und Theoretische Elektrotechnik, Universität Erlangen-Nürnberg, Cauerstrasse 7, D-8520 Erlangen, Germany Received: February 06 1992 Accepted: June 05 1992 Online ISSN: 1530-888X Print ISSN: 0899-7667 © 1993 Massachusetts Institute of Technology1993 Neural Computation (1993) 5 (1): 18–20. https://doi.org/10.1162/neco.1993.5.1.18 Article history Received: February 06 1992 Accepted: June 05 1992 Cite Icon Cite Permissions Share Icon Share Facebook Twitter LinkedIn Email Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation Ji-Nan Lin, Rolf Unbehauen; On the Realization of a Kolmogorov Network. Neural Comput 1993; 5 (1): 18–20. doi: https://doi.org/10.1162/neco.1993.5.1.18 Download citation file: Ris (Zotero) Reference Manager EasyBib Bookends Mendeley Papers EndNote RefWorks BibTex toolbar search Search Dropdown Menu toolbar search search input Search input auto suggest filter your search All ContentAll JournalsNeural Computation Search Advanced Search This content is only available as a PDF.

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3. Statistical Mechanics for a Network of Spiking Neurons

Leonid Kruglyak, William Bialek

https://doi.org/10.1162/neco.1993.5.1.21 | pp. 21-31

We show that a simple statistical mechanics model can capture the collective behavior of large networks of spiking neurons. Qualitative arguments suggest that regularly firing neurons should be described by a planar "spin" of unit length. We extract these spins from spike trains and then measure the interaction Hamiltonian using simulations of small clusters of cells. Correlations among spike trains obtained from simulations of large arrays of cells are in quantitative agreement with the predictions from these Hamiltonians. We comment on the novel computational abilities of these "XY networks."

4. Acetylcholine and Learning in a Cortical Associative Memory

Michael E. Hasselmo

https://doi.org/10.1162/neco.1993.5.1.32 | pp. 32-44

Implementing associative memory function in biologically realistic networks raises difficulties not dealt with in previous associative memory models. In particular, during learning of overlapping input patterns, recall of previously stored patterns can interfere with the learning of new patterns. Most associative memory models avoid this difficulty by ignoring the effect of previously modified connections during learning, thereby clamping activity to the patterns to be learned. Here I propose that the effects of acetylcholine in cortical structures may provide a neuropsychological mechanism for this clamping. Recent brain slice experiments have shown that acetylcholine selectively suppresses excitatory intrinsic fiber synaptic transmission within the olfactory cortex, while leaving excitatory afferent input unaffected. In a computational model of olfactory cortex, this selective suppression, applied during learning, prevents interference from previously stored patterns during the learning of new patterns. Analysis of the model shows that the amount of suppression necessary to prevent interference depends on cortical parameters such as inhibition and the threshold of synaptic modification, as well as input parameters such as the amount of overlap between the patterns being stored.

5. Convergent Algorithm for Sensory Receptive Field Development

Joseph J. Atick, Amanda Redlich

https://doi.org/10.1162/neco.1993.5.1.45 | pp. 45-60

An unsupervised developmental algorithm for linear maps is derived which reduces the pixel-entropy (using the measure introduced in previous work) at every update and thus removes pairwise correlations between pixels. Since the measure of pixel-entropy has only a global minimum the algorithm is guaranteed to converge to the minimum entropy map. Such optimal maps have recently been shown to possess cognitively desirable properties and are likely to be used by the nervous system to organize sensory information. The algorithm derived here turns out to be one proposed by Goodall for pairwise decorrelation.

It is biologically plausible since in a neural network implementation it requires only data available locally to a neuron. In training over ensembles of two-dimensional input signals with the same spatial power spectrum as natural scenes, networks develop output neurons with center-surround receptive fields similar to those of ganglion cells in the retina. Some technical issues pertinent to developmental algorithms of this sort, such as “symmetry fixing,” are also discussed.

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6. Three-Dimensional Object Recognition Using an Unsupervised BCM Network: The Usefulness of Distinguishing Features

Nathan Intrator, Joshua I. Gold

https://doi.org/10.1162/neco.1993.5.1.61 | pp. 61-74

We propose an object recognition scheme based on a method for feature extraction from gray level images that corresponds to recent statistical theory, called projection pursuit, and is derived from a biologically motivated feature extracting neuron. To evaluate the performance of this method we use a set of very detailed psychophysical three-dimensional object recognition experiments (Bülthoff and Edelman 1992).

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7. Complexity Optimized Data Clustering by Competitive Neural Networks

Joachim M. Buhmann, H. Kühnel

https://doi.org/10.1162/neco.1993.5.1.75 | pp. 75-88

Data clustering is a complex optimization problem with applications ranging from vision and speech processing to data transmission and data storage in technical as well as in biological systems. We discuss a clustering strategy that explicitly reflects the tradeoff between simplicity and precision of a data representation. The resulting clustering algorithm jointly optimizes distortion errors and complexity costs. A maximum entropy estimation of the clustering cost function yields an optimal number of clusters, their positions, and their cluster probabilities. Our approach establishes a unifying framework for different clustering methods like K-means clustering, fuzzy clustering, entropy constrained vector quantization, or topological feature maps and competitive neural networks.

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8. Clustering Data by Melting

Yiu-fai Wong

https://doi.org/10.1162/neco.1993.5.1.89 | pp. 89-104

We derive a new clustering algorithm based on information theory and statistical mechanics, which is the only algorithm that incorporates scale. It also introduces a new concept into clustering: cluster independence. The cluster centers correspond to the local minima of a thermodynamic free energy, which are identified as the fixed points of a one-parameter nonlinear map. The algorithm works by melting the system to produce a tree of clusters in the scale space. Melting is also insensitive to variability in cluster densities, cluster sizes, and ellipsoidal shapes and orientations. We tested the algorithm successfully on both simulated data and a Synthetic Aperture Radar image of an agricultural site with 12 attributes for crop identification.

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9. Coarse Coding Resource-Allocating Network

Gustavo Deco, Jürgen Ebmeyer

https://doi.org/10.1162/neco.1993.5.1.105 | pp. 105-114

In recent years localized receptive fields have been the subject of intensive research, due to their learning speed and efficient reconstruction of hypersurfaces. A very efficient implementation for such a network was proposed recently by Platt (1991). This resource-allocating network (RAN) allocates a new neuron whenever an unknown pattern is presented at its input layer. In this paper we introduce a new network architecture and learning paradigm. The aim of our approach is to incorporate "coarse coding" to the resource-allocating network. The network presented here provides for each input coordinate a separate layer, which consists of one-dimensional, locally tuned gaussian neurons. In the following layer multidimensional receptive fields are built by using pi-neurons. Linear neurons aggregate the outputs of the pi-neurons in order to approximate the required input-output mapping. The learning process follows the ideas of the resource-allocating network of Platt but due to the extended architecture of our network other improvements of the learning process had to be defined. Compared to the resource-allocating network a more compact network with comparable accuracy is obtained.

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10. Training Periodic Sequences Using Fourier Series Error Criterion

James A. Kottas

https://doi.org/10.1162/neco.1993.5.1.115 | pp. 115-131

Training a network to learn a set of periodic input/output sequences effectively makes the network learn a mapping between amplitudes and phases in Fourier space. The spectral backpropagation (SBP) training algorithm is a different way of doing this task. It measures the Fourier series components of the output error sequences and minimizes the total spectral energy as an adaptation criterion. This approach can train not only the weights but also time delays associated with the interconnects. Furthermore, the cells can have finite bandwidth via a first-order low-pass filter. Having adaptable time delays gives the SBP algorithm a powerful way to control the phase characteristics of the network.

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11. Generalization and Approximation Capabilities of Multilayer Networks

Yoshikane Takahashi

https://doi.org/10.1162/neco.1993.5.1.132 | pp. 132-139

This paper develops a theory for constructing 3-layered networks. The theory allows one to specify a finite discrete set of training data and a network structure (minimum intermediate units, synaptic weights and biases) that generalizes and approximates any given continuous mapping between sets of contours on a plane within any given permissible error.

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12. Statistical Theory of Learning Curves under Entropic Loss Criterion

■■■-■■■ ■■■■■, *Noboru Murata*

https://doi.org/10.1162/neco.1993.5.1.140 | pp. 140-153

The present paper elucidates a universal property of learning curves, which shows how the generalization error, training error, and the complexity of the underlying stochastic machine are related and how the behavior of a stochastic machine is improved as the number of training examples increases. The error is measured by the entropic loss. It is proved that the generalization error converges to H 0 , the entropy of the conditional distribution of the true machine, as H 0 + m * /(2t), while the training error converges as H 0 - m * /(2t), where t is the number of examples and m * shows the complexity of the network. When the model is faithful, implying that the true machine is in the model, m * is reduced to m, the number of modifiable parameters. This is a universal law because it holds for any regular machine irrespective of its structure under the maximum likelihood estimator. Similar relations are obtained for the Bayes and Gibbs learning algorithms. These learning curves show the relation among the accuracy of learning, the complexity of a model, and the number of training examples.

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13. Learning in the Recurrent Random Neural Network

Erol Gelenbe

https://doi.org/10.1162/neco.1993.5.1.154 | pp. 154-164

The capacity to learn from examples is one of the most desirable features of neural network models. We present a learning algorithm for the recurrent random network model (Gelenbe 1989, 1990) using gradient descent of a quadratic error function. The analytical properties of the model lead to a "backpropagation" type algorithm that requires the solution of a system of n linear and n nonlinear equations each time the n-neuron network "learns" a new input-output pair.

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14. Neural Networks and Nonlinear Adaptive Filtering: Unifying Concepts and New Algorithms

O. Nerrand, P. Roussel-Ragot, L. Personnaz, Gérard Dreyfus, Sylvie Marcos

https://doi.org/10.1162/neco.1993.5.2.165 | pp. 165-199

The paper proposes a general framework that encompasses the training of neural networks and the adaptation of filters. We show that neural networks can be considered as general nonlinear filters that can be trained adaptively, that is, that can undergo continual training with a possibly infinite number of time-ordered examples. We introduce the canonical form of a neural network. This canonical form permits a unified presentation of network architectures and of gradient-based training algorithms for both feedforward networks (transversal filters) and feedback networks (recursive filters). We show that several algorithms used classically in linear adaptive filtering, and some algorithms suggested by other authors for training neural networks, are special cases in a general classification of training algorithms for feedback networks.

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15. Fast Calculation of Synaptic Conductances

Rajagopal Srinivasan, Hillel J. Chiel

https://doi.org/10.1162/neco.1993.5.2.200 | pp. 200-204

The drawback of these solutions is that one must keep track of the times of occurrences of each spike that initiated the synaptic potentials, and recalculate each exponential in the summation at each time step. This

creates a large storage and computational overhead. Since both these equations represent the impulse response of a second-order differential equation, another approach is to numerically integrate additional differential equations for each synapse in the network (Wilson and Bower 1989). We have developed an improved method for computing synaptic conductances that separates equations 1 and 2 into two components: one that is a function of the current time of the simulation and one that accumulates the contributions of previous spike events to the synaptic conductance. We demonstrate that this method requires only the storage of two running sums and the time constants for each synapse, and that it is mathematically equivalent to equations 1 and 2. We will then demonstrate that it is also faster for a given level of precision than numerically integrating differential equations for each synapse. We will first describe our algorithm for equation 1, and then for equation 2.

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16. The Variance of Covariance Rules for Associative Matrix Memories and Reinforcement Learning

Peter Dayan, Terrence J. Sejnowski

https://doi.org/10.1162/neco.1993.5.2.205 | pp. 205-209

Hebbian synapses lie at the heart of most associative matrix memories (Kohonen 1987; Hinton and Anderson 1981) and are also biologically plausible (Brown et al. 1990; Baudry and Davis 1991). Their analytical and computational tractability make these memories the best understood form of distributed information storage. A variety of Hebbian algorithms for estimating the covariance between input and output patterns has been proposed. This note points out that one class of these involves stochastic estimation of the covariance, shows that the signal-to-noise ratios of the rules are governed by the variances of their estimates, and considers some parallels in reinforcement learning.

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17. Optimal Network Construction by Minimum Description Length

Gary D. Kendall, Trevor J. Hall

https://doi.org/10.1162/neco.1993.5.2.210 | pp. 210-212

March 01 1993 Optimal Network Construction by Minimum Description Length In Special Collection: CogNet Gary D. Kendall, Gary D. Kendall Department of Physics, King's College London, Strand, London WC2R 2LS, UK Search for other works by this author on: This Site Google Scholar Trevor J. Hall Trevor J. Hall Department of Physics, King's College London, Strand, London WC2R 2LS, UK Search for other works by this author on: This Site Google Scholar Author and Article Information Gary D. Kendall Department of Physics, King's College London, Strand, London WC2R 2LS, UK Trevor J. Hall Department of Physics, King's College London, Strand, London WC2R 2LS, UK Received: April 10 1992 Accepted: September 03 1992 Online ISSN: 1530-888X Print ISSN: 0899-7667 © 1993 Massachusetts Institute of Technology1993 Neural Computation (1993) 5 (2): 210–212. https://doi.org/10.1162/neco.1993.5.2.210 Article history Received: April 10 1992 Accepted: September 03 1992 Cite Icon Cite Permissions Share Icon Share Facebook Twitter LinkedIn MailTo Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation Gary D. Kendall, Trevor J. Hall; Optimal Network Construction by Minimum Description Length. Neural Comput 1993; 5 (2): 210–212. doi: https://doi.org/10.1162/neco.1993.5.2.210 Download citation file: Ris (Zotero) Reference Manager EasyBib Bookends Mendeley Papers EndNote RefWorks BibTex toolbar search Search Dropdown Menu toolbar

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18. A Neural Network Model of Inhibitory Information Processing in Aplysia

Diana E. J. Blazis, Thomas M. Fischer, Thomas Carew

https://doi.org/10.1162/neco.1993.5.2.213 | pp. 213-227

Recent cellular studies have revealed a novel form of inhibitory information processing in the siphon withdrawal reflex of the marine mollusc Aplysia: Motorneuronal output is significantly reduced by activity-dependent potentiation of recurrent inhibition within the siphon withdrawal network (Fischer and Carew 1991, 1993). This inhibitory modulation is mediated by two types of identified interneurons, L29s and L30s. In an effort to describe and analyze this and other forms of inhibitory information processing in Aplysia, and to compare it with similar processing in other nervous systems, we have constructed a neural network model that incorporates many empirically observed features of these interneurons. The model generates important aspects of the interactions of cells L29 and L30, and with no further modification, exhibits many network level phenomena that were not explicitly incorporated into the model.

19. Computational Diversity in a Formal Model of the Insect Olfactory Macroglomerulus

Christiane Linster, C. Masson, Michel Kerszberg, L. Personnaz, Gérard Dreyfus

https://doi.org/10.1162/neco.1993.5.2.228 | pp. 228-241

We present a model of the specialist olfactory system of selected moth species and the cockroach. The model is built in a semirandom fashion, constrained by biological (physiological and anatomical) data. We propose a classification of the response patterns of individual neurons, based on the temporal aspects of the observed responses. Among the observations made in our simulations a number relate to data about olfactory information processing reported in the literature; others may serve as predictions and as guidelines for further investigations. We discuss the effect of the stochastic parameters of the model on the observed model behavior and on the ability of the model to extract features of the input stimulation. We conclude that a formal network, built with random connectivity, can suffice to reproduce and to explain many aspects of olfactory information processing at the first level of the specialist olfactory system of insects.

20. Learning Competition and Cooperation

Sungzoon Cho, James A. Reggia

https://doi.org/10.1162/neco.1993.5.2.242 | pp. 242-259

Competitive activation mechanisms introduce competitive or inhibitory interactions between units through functional mechanisms instead of inhibitory connections. A unit receives input from another unit proportional to its own activation as well as to that of the sending unit and the connection strength between the two. This,

plus the finite output from any unit, induces competition among units that receive activation from the same unit. Here we present a backpropagation learning rule for use with competitive activation mechanisms and show empirically how this learning rule successfully trains networks to perform an exclusive-OR task and a diagnosis task. In particular, networks trained by this learning rule are found to outperform standard backpropagation networks with novel patterns in the diagnosis problem. The ability of competitive networks to bring about context-sensitive competition and cooperation among a set of units proved to be crucial in diagnosing multiple disorders.

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21. Constraints on Synchronizing Oscillator Networks

David Cairns, Roland Baddeley, Leslie S. Smith

https://doi.org/10.1162/neco.1993.5.2.260 | pp. 260-266

This paper investigates the constraints placed on some synchronized oscillator models by their underlying dynamics. Phase response graphs are used to determine the phase locking behaviors of three oscillator models. These results are compared with idealized phase response graphs for single phase and multiple phase systems. We find that all three oscillators studied are best suited to operate in a single phase system and that the requirements placed on oscillatory models for operation in a multiple phase system are not compatible with the underlying dynamics of oscillatory behavior for these types of oscillator models.

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22. Learning Mixture Models of Spatial Coherence

Suzanna Becker, Geoffrey E. Hinton

https://doi.org/10.1162/neco.1993.5.2.267 | pp. 267-277

We have previously described an unsupervised learning procedure that discovers spatially coherent properties of the world by maximizing the information that parameters extracted from different parts of the sensory input convey about some common underlying cause. When given random dot stereograms of curved surfaces, this procedure learns to extract surface depth because that is the property that is coherent across space. It also learns how to interpolate the depth at one location from the depths at nearby locations (Becker and Hinton 1992b). In this paper, we propose two new models that handle surfaces with discontinuities. The first model attempts to detect cases of discontinuities and reject them. The second model develops a mixture of expert interpolators. It learns to detect the locations of discontinuities and to invoke specialized, asymmetric interpolators that do not cross the discontinuities.

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23. Hints and the VC Dimension

Yaser S. Abu■Mostafa

https://doi.org/10.1162/neco.1993.5.2.278 | pp. 278-288

Learning from hints is a generalization of learning from examples that allows for a variety of information about the unknown function to be used in the learning process. In this paper, we use the VC dimension, an established tool for analyzing learning from examples, to analyze learning from hints. In particular, we show how the VC dimension is affected by the introduction of a hint. We also derive a new quantity that defines a VC dimension for the hint itself. This quantity is used to estimate the number of examples needed to

"absorb" the hint. We carry out the analysis for two types of hints, invariances and catalysts. We also describe how the same method can be applied to other types of hints.

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24. Redundancy Reduction as a Strategy for Unsupervised Learning

Amanda Redlich

https://doi.org/10.1162/neco.1993.5.2.289 | pp. 289-304

A redundancy reduction strategy, which can be applied in stages, is proposed as a way to learn as efficiently as possible the statistical properties of an ensemble of sensory messages. The method works best for inputs consisting of strongly correlated groups, that is features, with weaker statistical dependence between different features. This is the case for localized objects in an image or for words in a text. A local feature measure determining how much a single feature reduces the total redundancy is derived which turns out to depend only on the probability of the feature and of its components, but not on the statistical properties of any other features. The locality of this measure makes it ideal as the basis for a "neural" implementation of redundancy reduction, and an example of a very simple non-Hebbian algorithm is given. The effect of noise on learning redundancy is also discussed.

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25. Approximation and Radial-Basis-Function Networks

Jooyoung Park, Irwin W. Sandberg

https://doi.org/10.1162/neco.1993.5.2.305 | pp. 305-316

This paper concerns conditions for the approximation of functions in certain general spaces using radial-basis-function networks. It has been shown in recent papers that certain classes of radial-basis-function networks are broad enough for universal approximation. In this paper these results are considerably extended and sharpened.

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26. A Polynomial Time Algorithm for Generating Neural Networks for Pattern Classification: Its Stability Properties and Some Test Results

Somnath Mukhopadhyay, Asim Roy, Lark Sang Kim, Sandeep Govil

https://doi.org/10.1162/neco.1993.5.2.317 | pp. 317-330

Polynomial time training and network design are two major issues for the neural network community. A new algorithm has been developed that can learn in polynomial time and also design an appropriate network. The algorithm is for classification problems and uses linear programing models to design and train the network. This paper summarizes the new algorithm, proves its stability properties, and provides some computational results to demonstrate its potential.

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27. Neural Networks for Optimization Problems with Inequality Constraints: The Knapsack Problem

Mattias Ohlsson, Carsten Peterson, Bo Söderberg

https://doi.org/10.1162/neco.1993.5.2.331 | pp. 331-339

A strategy for finding approximate solutions to discrete optimization problems with inequality constraints using mean field neural networks is presented. The constraints x ≤ 0 are encoded by x■(x) terms in the energy function. A careful treatment of the mean field approximation for the self-coupling parts of the energy is crucial, and results in an essentially parameter-free algorithm. This methodology is extensively tested on the knapsack problem of size up to 10 3 items. The algorithm scales like NM for problems with N items and M constraints. Comparisons are made with an exact branch and bound algorithm when this is computationally possible (N ≤ 30). The quality of the neural network solutions consistently lies above 95% of the optimal ones at a significantly lower CPU expense. For the larger problem sizes the algorithm is compared with simulated annealing and a modified linear programming approach. For "nonhomogeneous" problems these produce good solutions, whereas for the more difficult "homogeneous" problems the neural approach is a winner with respect to solution quality and/or CPU time consumption. The approach is of course also applicable to other problems of similar structure, like set covering.

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28. A Model for Motor Endplate Morphogenesis: Diffusible Morphogens, Transmembrane Signaling, and Compartmentalized Gene Expression

Michel Kerszberg, Jean■Pierre Changeux

https://doi.org/10.1162/neco.1993.5.3.341 | pp. 341-358

A mathematical model for the formation and maintenance of synaptic contacts at the motor endplate is proposed. It is based on diffusion between sarcoplasmic nuclei of limiting amounts of a morphogen substance. The morphogen is postulated to act on genetic switch-like intranuclear units and to regulate positively both the transcription of its own gene and that of acetylcholine receptor (AChR) subunit genes. The efficacy of autoregulation is assumed to be depressed by electrical activity; while AChR genes transcription is enhanced by anterograde neural factors. Thus the model involves Turing's classical ingredients: autocatalysis and short range activation by the morphogen, and long range inhibition by electrical activity. Our predictions include: the stabilization of a single, transcriptionally active nucleus located in the central region of the developing muscle fiber (or myotube); the frequent occurrence of transcriptional activity in nuclei at the tendinous ends; and the onset, upon denervation of adult muscle, of transcription waves, starting from both the central site and the tendinous nuclei. In noninnervated fibers, the calculations show that spontaneous, irregular electrical activity leads to a variety of near-periodic spatial patterns of transcription; these are also predicted in innervated fibers when the depressing effect of electrical activity is weak, giving rise to the stabilization of multiple endplates as occurs in muscles with distributed innervation.

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29. Universal Approximation by Phase Series and Fixed-Weight Networks

N.E. Cotter, Peter R. Conwell

https://doi.org/10.1162/neco.1993.5.3.359 | pp. 359-362

In this note we show that weak (specified energy bound) universal approximation by neural networks is possible if variable synaptic weights are brought in as network inputs rather than being embedded in a network. We illustrate this idea with a Fourier series network that we transform into what we call a phase

series network. The transformation only increases the number of neurons by a factor of two.

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30. Backpropagation with Homotopy

Liping Yang, Wanzhen Yu

https://doi.org/10.1162/neco.1993.5.3.363 | pp. 363-366

When training a feedforward neural network with backpropagation (Rumelhart et al. 1986), local minima are always a problem because of the nonlinearity of the system. There have been several ways to attack this problem: for example, to restart the training by selecting a new initial point, to perform the preprocessing of the input data or the neural network. Here, we propose a method which is efficient in computation to avoid some local minima.

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31. Improving Rejection Performance on Handwritten Digits by Training with “Rubbish”

Jane Bromley, John S. Denker

https://doi.org/10.1162/neco.1993.5.3.367 | pp. 367-370

May 01 1993 Improving Rejection Performance on Handwritten Digits by Training with “Rubbish” In Special Collection: CogNet Jane Bromley, Jane Bromley AT&T Bell Laboratories, Holmdel, NJ 07733, USA Search for other works by this author on: This Site Google Scholar John S. Denker John S. Denker AT&T Bell Laboratories, Holmdel, NJ 07733, USA Search for other works by this author on: This Site Google Scholar Author and Article Information Jane Bromley AT&T Bell Laboratories, Holmdel, NJ 07733, USA John S. Denker AT&T Bell Laboratories, Holmdel, NJ 07733, USA Received: July 27 1992 Accepted: October 20 1992 Online ISSN: 1530-888X Print ISSN: 0899-7667 © 1993 Massachusetts Institute of Technology1993 Neural Computation (1993) 5 (3): 367–370. https://doi.org/10.1162/neco.1993.5.3.367 Article history Received: July 27 1992 Accepted: October 20 1992 Cite Icon Cite Permissions Share Icon Share Facebook Twitter LinkedIn MailTo Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation Jane Bromley, John S. Denker; Improving Rejection Performance on Handwritten Digits by Training with “Rubbish”. Neural Comput 1993; 5 (3): 367–370. doi: https://doi.org/10.1162/neco.1993.5.3.367 Download citation file: Ris (Zotero) Reference Manager EasyBib Bookends Mendeley Papers EndNote RefWorks BibTex toolbar search Search Dropdown Menu toolbar search search input Search input auto suggest filter your search All ContentAll JournalsNeural Computation Search Advanced Search This content is only available as a PDF. © 1993 Massachusetts Institute of Technology1993 Article PDF first page preview Close Modal You do not currently have access to this content.

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32. Vapnik-Chervonenkis Dimension Bounds for Two- and Three-Layer Networks

Peter L. Bartlett

https://doi.org/10.1162/neco.1993.5.3.371 | pp. 371-373

We show that the Vapnik-Chervonenkis dimension of the class of functions that can be computed by arbitrary two-layer or some completely connected three-layer threshold networks with real inputs is at least linear in the number of weights in the network. In Valiant's "probably approximately correct" learning framework, this implies that the number of random training examples necessary for learning in these networks is at least linear in the number of weights.

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33. A Neural Network for the Processing of Optic Flow from Ego-Motion in Man and Higher Mammals

Markus Lappe, Josef P. Rauschecker

https://doi.org/10.1162/neco.1993.5.3.374 | pp. 374-391

Interest in the processing of optic flow has increased recently in both the neurophysiological and the psychophysical communities. We have designed a neural network model of the visual motion pathway in higher mammals that detects the direction of heading from optic flow. The model is a neural implementation of the subspace algorithm introduced by Heeger and Jepson (1990). We have tested the network in simulations that are closely related to psychophysical and neurophysiological experiments and show that our results are consistent with recent data from both fields. The network reproduces some key properties of human ego-motion perception. At the same time, it produces neurons that are selective for different components of ego-motion flow fields, such as expansions and rotations. These properties are reminiscent of a subclass of neurons in cortical area MSTd, the triple-component neurons. We propose that the output of such neurons could be used to generate a computational map of heading directions in or beyond MST.

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34. Arbitrary Elastic Topologies and Ocular Dominance

Peter Dayan

https://doi.org/10.1162/neco.1993.5.3.392 | pp. 392-401

The elastic net, which has been used to produce accounts of the formation of topology-preserving maps and ocular dominance stripes (OD), embodies a nearest neighbor topology. A Hebbian account of OD is not so restricted—and indeed makes the prediction that the width of the stripes depends on the nature of the (more general) neighborhood relations. Elastic and Hebbian accounts have recently been unified, raising a question mark about their different determiners of stripe widths. This paper considers this issue, and demonstrates theoretically that it is possible to use more general topologies in the elastic net, including those effectively adopted in the Hebbian model.

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35. Neural Networks for Fingerprint Recognition

Pierre Baldi, Yves Chauvin

https://doi.org/10.1162/neco.1993.5.3.402 | pp. 402-418

After collecting a data base of fingerprint images, we design a neural network algorithm for fingerprint recognition. When presented with a pair of fingerprint images, the algorithm outputs an estimate of the probability that the two images originate from the same finger. In one experiment, the neural network is trained using a few hundred pairs of images and its performance is subsequently tested using several

thousand pairs of images originated from a subset of the database corresponding to 20 individuals. The error rate currently achieved is less than 0.5%. Additional results, extensions, and possible applications are also briefly discussed.

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36. Centered-Object Integrated Segmentation and Recognition of Overlapping Handprinted Characters

Gale L. Martin

https://doi.org/10.1162/neco.1993.5.3.419 | pp. 419-429

Visual object recognition is often conceived of as a final step in a visual processing system, First, physical information in the raw image is used to isolate and enhance to-be-recognized clumps and then each of the resulting preprocessed representations is fed into the recognizer. This general conception fails when there are no reliable physical cues for isolating the objects, such as when objects overlap. This paper describes an approach, called centered object integrated segmentation and recognition (COISR), for integrating object segmentation and recognition within a single neural network. The application is handprinted character recognition. The approach uses a backpropagation network that scans a field of characters and is trained to recognize whether it is centered over a single character or between characters. When it is centered over a character, the net classifies the character. The approach is tested on a dataset of handprinted digits and high accuracy rates are reported.

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37. Surface Interpolation Networks

Alex Pentland

https://doi.org/10.1162/neco.1993.5.3.430 | pp. 430-442

Orthogonal wavelets can be used as models for receptive fields in the human visual system. They may also be used to solve spatial interpolation problems formulated either as regularization or 2-D Kalman filtering. The solutions take the form of simple feedback networks, and only a few iterations are required for convergence.

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38. Combining Exploratory Projection Pursuit and Projection Pursuit Regression with Application to Neural Networks

Nathan Intrator

https://doi.org/10.1162/neco.1993.5.3.443 | pp. 443-455

We present a novel classification and regression method that combines exploratory projection pursuit (unsupervised training) with projection pursuit regression (supervised training), to yield a new family of cost/complexity penalty terms. Some improved generalization properties are demonstrated on real-world problems.

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39. A Simplified Gradient Algorithm for IIR Synapse Multilayer Perceptrons

Andrew D. Back, Ah Chung Tsoi
https://doi.org/10.1162/neco.1993.5.3.456 | pp. 456-462

A network architecture with a global feedforward local recurrent construction was presented recently as a new means of modeling nonlinear dynamic time series (Back and Tsoi 1991a). The training rule used was based on minimizing the least mean square (LMS) error and performed well, although the amount of memory required for large networks may become significant if a large number of feedback connections are used. In this note, a modified training algorithm based on a technique for linear filters is presented, simplifying the gradient calculations significantly. The memory requirements are reduced from O[n a (n a + n b)N s] to O[(2n a + n b)N s], where n a is the number of feedback delays, and N s is the total number of synapses. The new algorithm reduces the number of multiply-adds needed to train each synapse by n a at each time step. Simulations indicate that the algorithm has almost identical performance to the previous one.

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40. The Characteristics of the Convergence Time of Associative Neural Networks

Toshiaki Tanaka, Miki Yamada
https://doi.org/10.1162/neco.1993.5.3.463 | pp. 463-472

The authors have analyzed the dynamics of associative neural networks based on macroscopic state equations and have shown that both a layered associative net and an autocorrelation type net have the same convergence property: If a recalling process succeeds, the network converges very fast to one of the memorized patterns. But if a recalling process fails, it converges very slowly to a spurious state or does not converge. This property was also checked by computer simulations on a large scale (N = 1000) neural network. Moreover, it is shown that the convergence time for a successful recall is of order log(N). If this convergence time difference is used, execution time and memory can be saved and it can be determined whether a recalling process succeeds or fails without any additional procedure.

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41. Robustness in Multilayer Perceptrons

P. Kerlirzin, F. Vallet
https://doi.org/10.1162/neco.1993.5.3.473 | pp. 473-482

In this paper, we study the robustness of multilayer networks versus the destruction of neurons. We show that the classical backpropagation algorithm does not lead to optimal robustness and we propose a modified algorithm that improves this capability.

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42. Pattern Discrimination Using Feedforward Networks: A Benchmark Study of Scaling Behavior

Thorsteinn Rögnvaldsson
https://doi.org/10.1162/neco.1993.5.3.483 | pp. 483-491

The discrimination powers of multilayer perceptron (MLP) and learning vector quantization (LVQ) networks are compared for overlapping gaussian distributions. It is shown, both analytically and with Monte Carlo

studies, that the MLP network handles high-dimensional problems in a more efficient way than LVQ. This is mainly due to the sigmoidal form of the MLP transfer function, but also to the fact that the MLP uses hyperplanes more efficiently. Both algorithms are equally robust to limited training sets and the learning curves fall off like 1/M, where M is the training set size, which is compared to theoretical predictions from statistical estimates and Vapnik-Chervonenkis bounds.

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43. A Neural Network That Learns to Interpret Myocardial Planar Thallium Scintigrams

Charles Rosenberg, Jacob Erel, Henri Atlan

https://doi.org/10.1162/neco.1993.5.3.492 | pp. 492-502

The planar thallium-201 (201 Tl) myocardial perfusion scintigram is a widely used diagnostic technique for detecting and estimating the risk of coronary artery disease. Interpretation is currently based on visual scoring of myocardial defects combined with image quantitation and is known to have a significant subjective component. Neural networks learned to interpret thallium scintigrams as determined by both individual and multiple (consensus) expert ratings. Four different types of networks were explored: single-layer, two-layer backpropagation (BP), BP with weight smoothing, and two-layer radial basis function (RBF). The RBF network was found to yield the best performance (94.8% generalization by region) and compares favorably with human experts. We conclude that this network is a valuable clinical tool that can be used as a reference "diagnostic support system" to help reduce inter- and intraobserver variability. This system is now being further developed to include other variables that are expected to improve the final clinical diagnosis.

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44. The Use of Neural Networks in High-Energy Physics

B. Denby

https://doi.org/10.1162/neco.1993.5.4.505 | pp. 505-549

In the past few years a wide variety of applications of neural networks to pattern recognition in experimental high-energy physics has appeared. The neural network solutions are in general of high quality, and, in a number of cases, are superior to those obtained using "traditional" methods. But neural networks are of particular interest in high-energy physics for another reason as well: much of the pattern recognition must be performed online, that is, in a few microseconds or less. The inherent parallelism of neural network algorithms, and the ability to implement them as very fast hardware devices, may make them an ideal technology for this application.

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45. Stimulus-Dependent Synchronization of Neuronal Assemblies

Eric R. Grannan, David Kleinfeld, Haim Sompolinsky

https://doi.org/10.1162/neco.1993.5.4.550 | pp. 550-569

We study theoretically how an interaction between assemblies of neuronal oscillators can be modulated by the pattern of external stimuli. It is shown that spatial variations in the stimuli can control the magnitude and phase of the synchronization between the output of neurons with different receptive fields. This modulation

emerges from cooperative dynamics in the network, without the need for specialized, activity-dependent synapses. Our results further suggest that the modulation of neuronal interactions by extended features of a stimulus may give rise to complex spatiotemporal fluctuations in the phases of neuronal oscillations.

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46. Dynamics of Populations of Integrate-and-Fire Neurons, Partial Synchronization and Memory

Marius Usher, Heinz Georg Schuster, Ernst Niebur

https://doi.org/10.1162/neco.1993.5.4.570 | pp. 570-586

We study the dynamics of completely connected populations of refractory integrate-and-fire neurons in the presence of noise. Solving the master equation based on a mean-field approach, and by computer simulations, we find sustained states of activity that correspond to fixed points and show that for the same value of external input, the system has one or two attractors. The dynamic behavior of the population under the influence of external input and noise manifests hysteresis effects that might have a functional role for memory. The temporal dynamics at higher temporal resolution, finer than the transmission delay times and the refractory period, are characterized by synchronized activity of subpopulations. The global activity of the population shows aperiodic oscillations analogous to experimentally found field potentials.

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47. The Effects of Cell Duplication and Noise in a Pattern Generating Network

Catherine H. Ferrar, Thelma L. Williams, Graham Bowtell

https://doi.org/10.1162/neco.1993.5.4.587 | pp. 587-596

Stability against stochastic variation is an important property for biological systems. This paper investigates the robustness of the rhythmic activity produced by a model of the segmental rhythm generator for locomotion in the lamprey, by introducing stochastic properties into the network. In addition, since neuronal models for vertebrate systems often use a single neuron to represent a large class of cells, this paper explores one of the consequences of such reduction, by investigating the effects of duplicating all the cells of the network on its stability against stochastic variation. We have found the basic model network to be very stable, and have found that this stability is increased by doubling the number of cells in the network.

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48. Emergence of Position-Independent Detectors of Sense of Rotation and Dilation with Hebbian Learning: An Analysis

Kechen Zhang, Martin I. Sereno, Margaret E. Sereno

https://doi.org/10.1162/neco.1993.5.4.597 | pp. 597-612

We previously demonstrated that it is possible to learn position-independent responses to rotation and dilation by filtering rotations and dilations with different centers through an input layer with MT-like speed and direction tuning curves and connecting them to an MST-like layer with simple Hebbian synapses (Sereno and Sereno 1991). By analyzing an idealized version of the network with broader, sinusoidal direction-tuning and linear speed-tuning, we show analytically that a Hebb rule trained with arbitrary rotation, dilation/contraction, and translation velocity fields yields units with weight fields that are a rotation plus a dilation or contraction field, and whose responses to a rotating or dilating/contracting disk are exactly

position independent. Differences between the performance of this idealized model and our original model (and real MST neurons) are discussed.

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49. Improving Generalization for Temporal Difference Learning: The Successor Representation

Peter Dayan

https://doi.org/10.1162/neco.1993.5.4.613 | pp. 613-624

Estimation of returns over time, the focus of temporal difference (TD) algorithms, imposes particular constraints on good function approximators or representations. Appropriate generalization between states is determined by how similar their successors are, and representations should follow suit. This paper shows how TD machinery can be used to learn such representations, and illustrates, using a navigation task, the appropriately distributed nature of the result.

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50. Discovering Predictable Classifications

Jürgen Schmidhuber, D. Prelinger

https://doi.org/10.1162/neco.1993.5.4.625 | pp. 625-635

Prediction problems are among the most common learning problems for neural networks (e.g., in the context of time series prediction, control, etc.). With many such problems, however, perfect prediction is inherently impossible. For such cases we present novel unsupervised systems that learn to classify patterns such that the classifications are predictable while still being as specific as possible. The approach can be related to the IMAX method of Becker and Hinton (1989) and Zemel and Hinton (1991). Experiments include a binary stereo task proposed by Becker and Hinton, which can be solved more readily by our system.

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51. A Kinetic Model of Short- and Long-Term Potentiation

Michele Migliore, G.F. Ayala

https://doi.org/10.1162/neco.1993.5.4.636 | pp. 636-647

We present a kinetic model that can account for several experimental findings on short- and long-term potentiation (STP and LTP) and their pharmacological modulation. The model, which is consistent with Hebb's postulate, uses the hypothesis that part of the origin of LTP may be a consequence of an increased release of neurotransmitter due to a retrograde signal. The operation of the model is expressed by a set of irreversible reactions, each of which should be thought of as equivalent to a set of more complex reactions. We show that a retrograde signal alone is not sufficient to maintain LTP unless long-term change of the rate constant of some of the reactions is caused by high-frequency stimulation. Pharmacological manipulation of LTP is interpreted as modifications of the rate constants of one or more of the reactions that express a given mechanism. The model, because of its simplicity, can be useful to test more specific mechanisms by expanding one or more reactions as suggested by new experimental evidence.

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52. Artificial Dendritic Trees

John G. Elias

https://doi.org/10.1162/neco.1993.5.4.648 | pp. 648-664

The electronic architecture and dynamic signal processing capabilities of an artificial dendritic tree that can be used to process and classify dynamic signals is described. The electrical circuit architecture is modeled after neurons that have spatially extensive dendritic trees. The artificial dendritic tree is a hybrid VLSI circuit and is sensitive to both temporal and spatial signal characteristics. It does not use the conventional neural network concept of weights, and as such it does not use multipliers, adders, look-up-tables, microprocessors, or other complex computational units to process signals. The weights of conventional neural networks, which take the form of numerical, resistive, voltage, or current values, but do not have any spatial or temporal content, are replaced with connections whose spatial location have both a temporal and scaling significance.

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53. Patterns of Local Connectivity in the Neocortex

Andrew J. Nicoll, Colin Blakemore

https://doi.org/10.1162/neco.1993.5.5.665 | pp. 665-680

Dual intracellular recording of nearby pairs of pyramidal cells in slices of rat visual cortex has shown that there are significant differences in functional connectivity between the superficial and deep layers (Mason et al. 1991; Nicoll and Blakemore 1993). For pairs of cells no farther than 300 μm apart, synaptic connections between layer 2/3 pyramidal neurons were individually weaker (median peak amplitude, A, of single-fiber excitatory postsynaptic potentials, EPSPs, = 0.4 mV) but more frequent (connection probability, p = 0.087) than those between layer 5 pyramidal neurons (mean A = 0.8 mV, p < 0.015). Taken in combination with plausible estimates of the density of pyramidal cells, the total numbers of synapses on them and the number of synapses formed on their intracortical axons, the present analysis of the above data suggests that roughly 70% of the excitatory synapses on any layer 2/3 pyramid, but fewer than 1% of those on a layer 5 pyramidal neuron, are derived from neighboring pyramidal neurons in its near vicinity. Even assuming very extreme values for some parameters, chosen to erode this difference, the calculated proportion of "local synapses" for layer 5 pyramids was always markedly lower than for layer 2/3 pyramidal neurons. These results imply that local excitatory connections are much more likely to provide significant "intracortical amplification" of afferent signals in layer 2/3 than in layer 5 of rat visual cortex.

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54. Sensitivity of Synaptic Plasticity to the Ca²⁺ Permeability of NMDA Channels: A Model of Long-Term Potentiation in Hippocampal Neurons

Erik De Schutter, James M. Bower

https://doi.org/10.1162/neco.1993.5.5.681 | pp. 681-694

We have examined a model by Holmes and Levy (1990) of the induction of associative long-term potentiation (LTP) by a rise in the free Ca 2+ concentration ([Ca 2+]) after synaptic activation of dendritic spines. The previously reported amplification of the change in [Ca 2+] caused by coactivation of several synapses was found to be quite sensitive to changes in the permeability of the N-methyl-D-aspartate

(NMDA) receptor channels to Ca 2+ . Varying this parameter indicated that maximum amplification is obtained at values that are close to Ca 2+ permeabilities reported in the literature. However, amplification failed if permeability is reduced by more than 50%. We also found that the maximum free [Ca 2+] reached in an individual spine during synaptic coactivation of several spines depended on the location of that spine on the dendritic tree. Distal spines attained a higher [Ca 2+] than proximal ones, with differences of up to 80%. The implications of this result for the uniformity of induction of associative LTP in spines in different regions of the dendrite are discussed.

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55. Models of Perceptual Learning in Vernier Hyperacuity

Yair Weiss, Shimon Edelman, Manfred Fahle

https://doi.org/10.1162/neco.1993.5.5.695 | pp. 695-718

Performance of human subjects in a wide variety of early visual processing tasks improves with practice. HyperBF networks (Poggio and Girosi 1990) constitute a mathematically well-founded framework for understanding such improvement in performance, or perceptual learning, in the class of tasks known as visual hyperacuity. The present article concentrates on two issues raised by the recent psychophysical and computational findings reported in Poggio et al. (1992b) and Fahle and Edelman (1992). First, we develop a biologically plausible extension of the HyperBF model that takes into account basic features of the functional architecture of early vision. Second, we explore various learning modes that can coexist within the HyperBF framework and focus on two unsupervised learning rules that may be involved in hyperacuity learning. Finally, we report results of psychophysical experiments that are consistent with the hypothesis that activity-dependent presynaptic amplification may be involved in perceptual learning in hyperacuity.

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56. Learning to Generalize from Single Examples in the Dynamic Link Architecture

Wolfgang Konen, Christoph von der Malsburg

https://doi.org/10.1162/neco.1993.5.5.719 | pp. 719-735

A large attraction of neural systems lies in their promise of replacing programming by learning. A problem with many current neural models is that with realistically large input patterns learning time explodes. This is a problem inherent in a notion of learning that is based almost entirely on statistical estimation. We propose here a different learning style where significant relations in the input pattern are recognized and expressed by the unsupervised self-organization of dynamic links. The power of this mechanism is due to the very general a priori principle of conservation of topological structure. We demonstrate that style with a system that learns to classify mirror symmetric pixel patterns from single examples.

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57. Neural Network Modeling of Memory Deterioration in Alzheimer's Disease

D. Horn, Eytan Ruppin, Marius Usher, Michael Herrmann

https://doi.org/10.1162/neco.1993.5.5.736 | pp. 736-749

The clinical course of Alzheimer's disease (AD) is generally characterized by progressive gradual deterioration, although large clinical variability exists. Motivated by the recent quantitative reports of synaptic

changes in AD, we use a neural network model to investigate how the interplay between synaptic deletion and compensation determines the pattern of memory deterioration, a clinical hallmark of AD. Within the model we show that the deterioration of memory retrieval due to synaptic deletion can be much delayed by multiplying all the remaining synaptic weights by a common factor, which keeps the average input to each neuron at the same level. This parallels the experimental observation that the total synaptic area per unit volume (TSA) is initially preserved when synaptic deletion occurs. By using different dependencies of the compensatory factor on the amount of synaptic deletion one can define various compensation strategies, which can account for the observed variation in the severity and progression rate of AD.

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58. Supervised Factorial Learning

Amanda Redlich

https://doi.org/10.1162/neco.1993.5.5.750 | pp. 750-766

Factorial learning, finding a statistically independent representation of a sensory “image”—a factorial code—is applied here to solve multilayer supervised learning problems that have traditionally required backpropagation. This lends support to Barlow's argument for factorial sensory processing, by demonstrating how it can solve actual pattern recognition problems. Two techniques for supervised factorial learning are explored, one of which gives a novel distributed solution requiring only positive examples. Also, a new nonlinear technique for factorial learning is introduced that uses neural networks based on almost reversible cellular automata. Due to the special functional connectivity of these networks—which resemble some biological microcircuits—learning requires only simple local algorithms. Also, supervised factorial learning is shown to be a viable alternative to backpropagation. One significant advantage is the existence of a measure for the performance of intermediate learning stages.

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59. On Learning Perceptrons with Binary Weights

Mostefa Golea, Mario Marchand

https://doi.org/10.1162/neco.1993.5.5.767 | pp. 767-782

We present an algorithm that PAC learns any perceptron with binary weights and arbitrary threshold under the family of product distributions. The sample complexity of this algorithm is of O[(n/ε) 4 ln(n/δ)] and its running time increases only linearly with the number of training examples. The algorithm does not try to find an hypothesis that agrees with all of the training examples; rather, it constructs a binary perceptron based on various probabilistic estimates obtained from the training examples. We show that, under the restricted case of the uniform distribution and zero threshold, the algorithm reduces to the well known clipped Hebb rule. We calculate exactly the average generalization rate (i.e., the learning curve) of the algorithm, under the uniform distribution, in the limit of an infinite number of dimensions. We find that the error rate decreases exponentially as a function of the number of training examples. Hence, the average case analysis gives a sample complexity of O[n ln(1/ε)], a large improvement over the PAC learning analysis. The analytical expression of the learning curve is in excellent agreement with the extensive numerical simulations. In addition, the algorithm is very robust with respect to classification noise.

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60. Construction of Minimal n-2-n Encoders for Any n

Dhananjay S. Phatak, Hyunjun Choi, Israel Koren
https://doi.org/10.1162/neco.1993.5.5.783 | pp. 783-794

The encoding problem (Rumelhart and McClelland 1986) is an important canonical problem. It has been widely used as a benchmark. Here, we have analytically derived minimal-sized nets necessary and sufficient to solve encoding problems of arbitrary size. The proofs are constructive: we construct n-2-n encoders and show that two hidden units are also necessary for n > 2. Moreover, the geometric approach employed is general and has much wider applications. For example, this method has also helped us derive lower bounds on the redundancy necessary for achieving complete fault tolerance (Phatak and Koren 1992a,b).

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61. Recurrent and Feedforward Polynomial Modeling of Coupled Time Series

Vicente López, Ramón Huerta, José R. Dorronsoro
https://doi.org/10.1162/neco.1993.5.5.795 | pp. 795-811

We present two methods for the prediction of coupled time series. The first one is based on modeling the series by a dynamic system with a polynomial format. This method can be formulated in terms of learning in a recurrent network, for which we give a computationally effective algorithm. The second method is a purely feedforward σ-π network procedure whose architecture derives from the recurrence relations for the derivatives of the trajectories of a Ricatti format dynamic system. It can also be used for the modeling of discrete series in terms of nonlinear mappings. Both methods have been tested successfully against chaotic series.

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62. Attraction Radii in Binary Hopfield Nets are Hard to Compute

Patrik Floréen, Pekka Orponen
https://doi.org/10.1162/neco.1993.5.5.812 | pp. 812-821

We prove that it is an NP-hard problem to determine the attraction radius of a stable vector in a binary Hopfield memory network, and even that the attraction radius is hard to approximate. Under synchronous updating, the problems are already NP-hard for two-step attraction radii; direct (one-step) attraction radii can be computed in polynomial time.

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63. Analysis of Neuron Models with Dynamically Regulated Conductances

L. F. Abbott, Gwendal Le Masson
https://doi.org/10.1162/neco.1993.5.6.823 | pp. 823-842

We analyze neuron models in which the maximal conductances of membrane currents are slowly varying dynamic variables regulated by the intracellular calcium concentration. These models allow us to study possible activity-dependent effects arising from processes that maintain and modify membrane channels in real neurons. Regulated model neurons maintain a constant average level of activity over a wide range of conditions by appropriately adjusting their conductances. The intracellular calcium concentration acts as a feedback element linking maximal conductances to electrical activity. The resulting plasticity of intrinsic characteristics has important implications for network behavior. We first study a simple two-conductance model, then introduce techniques that allow us to analyze dynamic regulation with an arbitrary number of

conductances, and finally illustrate this method by studying a seven-conductance model. We conclude with an analysis of spontaneous differentiation of identical model neurons in a two-cell network.

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64. Limitations of the Hodgkin-Huxley Formalism: Effects of Single Channel Kinetics on Transmembrane Voltage Dynamics

Adam F. Strassberg, Louis J. DeFelice

https://doi.org/10.1162/neco.1993.5.6.843 | pp. 843-855

A standard membrane model, based on the continuous deterministic Hodgkin-Huxley equations, is compared to an alternative membrane model, based on discrete stochastic ion channel populations represented through Markov processes. Simulations explore the relationship between these two levels of description: the behavior predicted by the macroscopic membrane currents versus the behavior predicted by their microscopic ion channels. Discussion considers the extent to which these random events underlying neural signals mediate random events in neural computation.

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65. Two-Dimensional Motion Perception in Flies

Alexander Borst, Martin Egelhaaf, H. Sebastian Seung

https://doi.org/10.1162/neco.1993.5.6.856 | pp. 856-868

We study two-dimensional motion perception in flies using a semicircular visual stimulus. Measurements of both the H1-neuron and the optomotor response are consistent with a simple model supposing spatial integration of the outputs of correlation-type motion detectors. In both experiment and model, there is substantial H1 and horizontal (yaw) optomotor response to purely vertical motion of the stimulus. We conclude that the fly's optomotor response to a two-dimensional pattern, depending on its structure, may deviate considerably from the direction of pattern motion.

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66. Neural Representation of Space Using Sinusoidal Arrays

David S. Touretzky, A. David Redish, Hank S. Wan

https://doi.org/10.1162/neco.1993.5.6.869 | pp. 869-884

O'Keefe (1991) has proposed that spatial information in rats might be represented as phasors: phase and amplitude of a sine wave encoding angle and distance to a landmark. We describe computer simulations showing that operations on phasors can be efficiently realized by arrays of spiking neurons that recode the temporal dimension of the sine wave spatially. Some cells in motor and parietal cortex exhibit response properties compatible with this proposal.

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67. Fast Recognition of Noisy Digits

Jeffrey N. Kidder, Daniel Seligson

https://doi.org/10.1162/neco.1993.5.6.885 | pp. 885-892

We describe a hardware solution to a high-speed optical character recognition (OCR) problem. Noisy 15 × 10 binary images of machine written digits were processed and applied as input to Intel's Electrically Trainable Analog Neural Network (ETANN). In software simulation, we trained an 80 × 54 × 10 feedforward network using a modified version of backprop. We then downloaded the synaptic weights of the trained network to ETANN and tweaked them to account for differences between the simulation and the chip itself. The best recognition error rate was 0.9% in hardware with a 3.7% rejection rate on a 1000-character test set.

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68. Local Algorithms for Pattern Recognition and Dependencies Estimation

Vladimir Vapnik, Léon Bottou

https://doi.org/10.1162/neco.1993.5.6.893 | pp. 893-909

In previous publications (Bottou and Vapnik 1992; Vapnik 1992) we described local learning algorithms, which result in performance improvements for real problems. We present here the theoretical framework on which these algorithms are based. First, we present a new statement of certain learning problems, namely the local risk minimization. We review the basic results of the uniform convergence theory of learning, and extend these results to local risk minimization. We also extend the structural risk minimization principle for both pattern recognition problems and regression problems. This extended induction principle is the basis for a new class of algorithms.

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69. On the Geometry of Feedforward Neural Network Error Surfaces

An Mei Chen, Haw-minn Lu, Robert Hecht-Nielsen

https://doi.org/10.1162/neco.1993.5.6.910 | pp. 910-927

Many feedforward neural network architectures have the property that their overall input-output function is unchanged by certain weight permutations and sign flips. In this paper, the geometric structure of these equioutput weight space transformations is explored for the case of multilayer perceptron networks with tanh activation functions (similar results hold for many other types of neural networks). It is shown that these transformations form an algebraic group isomorphic to a direct product of Weyl groups. Results concerning the root spaces of the Lie algebras associated with these Weyl groups are then used to derive sets of simple equations for minimal sufficient search sets in weight space. These sets, which take the geometric forms of a wedge and a cone, occupy only a minute fraction of the volume of weight space. A separate analysis shows that large numbers of copies of a network performance function optimum weight vector are created by the action of the equioutput transformation group and that these copies all lie on the same sphere. Some implications of these results for learning are discussed.

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70. Rational Function Neural Network

Henry Leung, S. Haykin

https://doi.org/10.1162/neco.1993.5.6.928 | pp. 928-938

In this paper we observe that a particular class of rational function (RF) approximations may be viewed as feedforward networks. Like the radial basis function (RBF) network, the training of the RF network may be

performed using a linear adaptive filtering algorithm. We illustrate the application of the RF network by considering two nonlinear signal processing problems. The first problem concerns the one-step prediction of a time series consisting of a pair of complex sinusoid in the presence of colored non-gaussian noise. Simulated data were used for this problem. In the second problem, we use the RF network to build a nonlinear dynamic model of sea clutter (radar backscattering from a sea surface); here, real-life data were used for the study.

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71. On an Unsupervised Learning Rule for Scalar Quantization following the Maximum Entropy Principle

Marc M. Van Hulle, Dominique Martinez

https://doi.org/10.1162/neco.1993.5.6.939 | pp. 939-953

A novel unsupervised learning rule, called Boundary Adaptation Rule (BAR), is introduced for scalar quantization. It is shown that the rule maximizes information-theoretic entropy and thus yields equiprobable quantizations of univariate probability density functions. It is shown by simulations that BAR outperforms other unsupervised competitive learning rules in generating equiprobable quantizations. It is also shown that our rule can do better or worse than the Lloyd I algorithm in minimizing average mean square error, depending on the input distribution. Finally, an application to adaptive non-uniform analog to digital (A/D) conversion is considered.

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72. A Function Estimation Approach to Sequential Learning with Neural Networks

Visakan Kadirkamanathan, Mahesan Niranjan

https://doi.org/10.1162/neco.1993.5.6.954 | pp. 954-975

In this paper, we investigate the problem of optimal sequential learning, viewed as a problem of estimating an underlying function sequentially rather than estimating a set of parameters of the neural network. First, we arrive at a suboptimal solution to the sequential estimate that can be mapped by a growing gaussian radial basis function (GaRBF) network. This network adds hidden units for each observation. The function space approach in which the estimates are represented as vectors in a function space is used in developing a growth criterion to limit its growth. A simplification of the criterion leads to two joint criteria on the distance of the present pattern and the existing unit centers in the input space and on the approximation error of the network for the given observation to be satisfied together. This network is similar to the resource allocating network (RAN) (Platt 1991a) and hence RAN can be interpreted from a function space approach to sequential learning. Second, we present an enhancement to the RAN. The RAN either allocates a new unit based on the novelty of an observation or adapts the network parameters by the LMS algorithm. The function space interpretation of the RAN lends itself to an enhancement of the RAN in which the extended Kalman filter (EKF) algorithm is used in place of the LMS algorithm. The performance of the RAN and the enhanced network are compared in the experimental tasks of function approximation and time-series prediction demonstrating the superior performance of the enhanced network with fewer number of hidden units. The approach adopted here has led us toward the minimal network required for a sequential learning problem.

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73. Learning Finite State Machines With Self-Clustering Recurrent Networks

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Recent work has shown that recurrent neural networks have the ability to learn finite state automata from examples. In particular, networks using second-order units have been successful at this task. In studying the performance and learning behavior of such networks we have found that the second-order network model attempts to form clusters in activation space as its internal representation of states. However, these learned states become unstable as longer and longer test input strings are presented to the network. In essence, the network “forgets” where the individual states are in activation space. In this paper we propose a new method to force such a network to learn stable states by introducing discretization into the network and using a pseudo-gradient learning rule to perform training. The essence of the learning rule is that in doing gradient descent, it makes use of the gradient of a sigmoid function as a heuristic hint in place of that of the hard-limiting function, while still using the discretized value in the feedback update path. The new structure uses isolated points in activation space instead of vague clusters as its internal representation of states. It is shown to have similar capabilities in learning finite state automata as the original network, but without the instability problem. The proposed pseudo-gradient learning rule may also be used as a basis for training other types of networks that have hard-limiting threshold activation functions.

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